Homework #1

This is the homework for week 2. The due date is 11:59 p.m., September 22. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (30 points) i. Suppose random variable X satisfies Possion distribution with parameter $\lambda > 0$, i.e., the probability mass function

$$Prob(X = k) = \frac{\lambda^k}{k!}\exp(-\lambda), \text{ where } k = 0, 1, \cdots,$$

Calculate $\mathbf{E}X$, $Var(X)$, and its moment generating function $H(t) = \mathbf{E}\exp(tX)$, where $t \in \mathbf{R}$.

ii. Denote $F(a)$ to be the cumulative distribution function of X, prove that $F(a) \geq 1 - \frac{\lambda}{a}$ for any $a > 0$.

iii. Now suppose random variable Y satisfies uniform distribution on $[0, 1]$, i.e., its density $f(t) = 1$ on $[0, 1]$. Derive the mean $\mathbf{E}Y$, the variance $Var(Y)$, the skewness γ, and the Kurtosis $Kurt(Y)$.

iv. Suppose $Z_1, Z_2, \cdots, Z_n$ are i.i.d. (independent and identical distributed) random variables, prove that

$$Var(\sum_{i=1}^{n} Z_i) = \sum_{i=1}^{n} Var(Z_i).$$

Hint: you can use this fact when you prove this question: If $Z_1, Z_2, \cdots, Z_n$ are independent, then for any function h, $h(Z_1, \cdots, Z_k)$ and Z_{k+1} is also independent.

(**Question2):** (20 points)

(Hint, see problem 5.3–8. in the textbook) Suppose two independent claims X and Y are made on two insured homes, where each claim has probability density function

$$f(x) = \frac{4}{x^5}, 1 < x.$$

Find the expected value and the variance of the larger claim $\max(X, Y)$.

(**Question3):** (30 points)

Suppose X and Y are independent random variables with mean $\mathbf{E}X = \mathbf{E}Y = \mu \neq 0$ and variance $Var(X) = Var(Y) = \sigma^2 > 0$. Find the correlation coefficient

$$Corr(XY, Y).$$

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. Generate 500 i.i.d. random variables $X_1, \cdots, X_{500}$ with uniform distribution on $[0, 1]$.

ii. Calculate and record the sample mean, sample variace, sample skewness, and sample Kurtosis (See the "Estimator" column of Table 1 of the lecture notes "probability review").

iii. Repeat experiment for 300 times, plot four histograms of the sample values. Make comparisons between the sample values and the true values.